Step 7 Duration-Aware Variable-Window Report

Overview

Metric	Value
Included participants	36
Included questions	30
Included trials	1075
Mean variable-window duration (s)	90.01
Primary condition p-value	0.102262
Step 6 benchmark p-value	0.052102

Duration diagnostics. Concrete mean=89.23s (median=80.66s); Abstract mean=91.12s (median=84.80s). Response time vs total word count: r=0.285, p=0.000000; Response time vs sentence count: r=0.151, p=0.000001; Response time vs sentence length: r=0.005, p=0.872070; Response time vs ENEM item correctness: r=-0.161, p=0.000000.

Warnings and QC. MODEL_PRIMARY_FRONT_VARIABLE_WINDOW: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_FRONT_SENTENCECOUNT: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_FRONT_SENTENCELENGTH: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_BACK_VARIABLE_WINDO Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_DISSOCIATION_VARIABLE_WINDOW: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_FRONT_LAG3: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_FRONT_LAG4: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_FRONT_LAG5: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_FRONT_CAP16: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.; MODEL_FRONT_RT_SENSITIVITY: Model type=MixedLM; optimizer=lbfgs; fallback_used=False; warnings=lbfgs: The MLE may be on the boundary of the parameter space.

Primary Front ROI Model

Metric	Value
Model type	MixedLM
Included participants	36
Included questions	30
Included trials	1075
Mean variable-window duration (s)	90.01
Median variable-window duration (s)	82.80
Condition beta	3.629e-08
Condition SE	2.221e-08
Condition statistic	1.634
Condition p-value	0.102262
Condition 95% CI	[-7.240e-09, 7.982e-08]
Word-count beta	-4.626e-08
ENEM-correctness beta	3.424e-09
Participant random-intercept variance	7.623e-15
Question random-intercept variance	6.306e-14

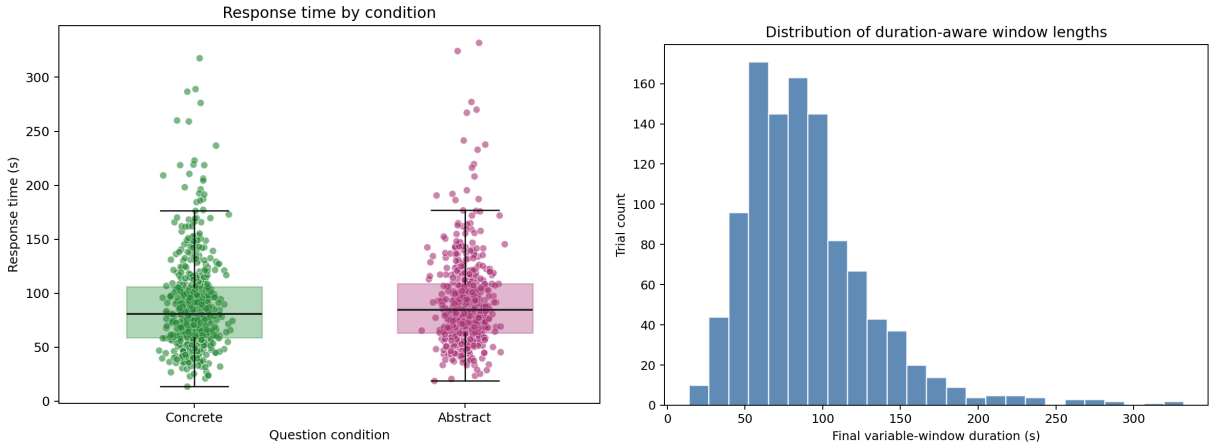

Figure 1: Response-time distribution by condition and the distribution of final duration-aware window lengths.

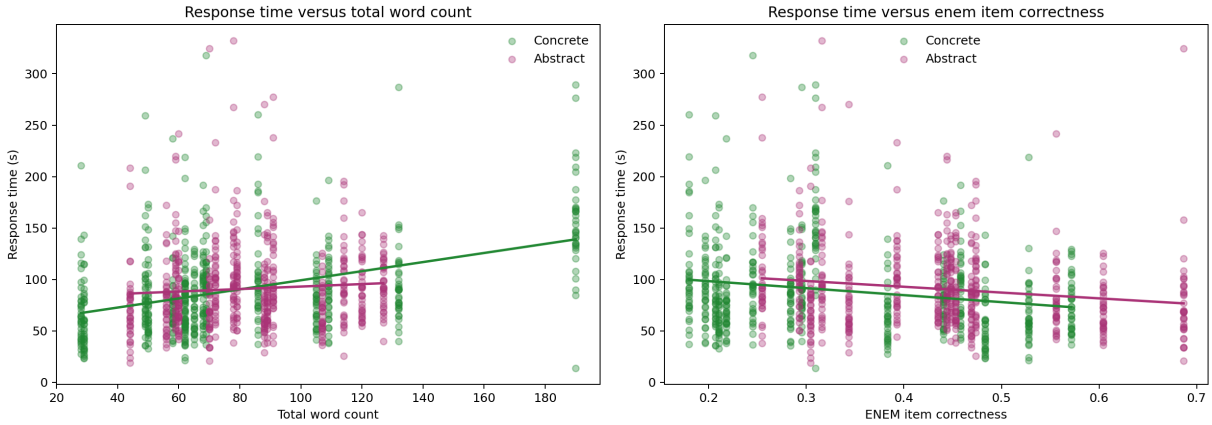

Figure 2: Response time plotted against total word count and ENEM item correctness.

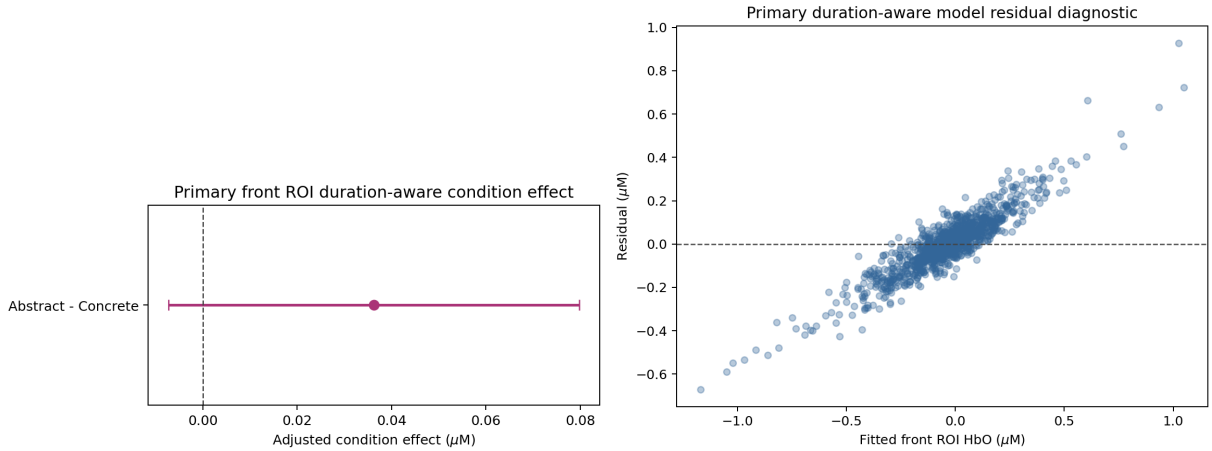

Figure 3: Primary duration-aware front ROI condition effect and residual diagnostic.

Secondary and Sensitivity Models

Model	Cond. beta	Cond. p	Mean dur. (s)	Fallback
Alternative front ROI adjusted for sentence count and ENEM correctness	1.195e-08	0.584651	90.01	no
Alternative front ROI adjusted for sentence length and ENEM correctness	1.800e-08	0.419344	90.01	no
Secondary back ROI adjusted for log word count and ENEM correctness	1.532e-08	0.372698	90.01	no
Secondary dissociation adjusted for log word count and ENEM correctness	2.100e-08	0.318036	90.01	no
Lag sensitivity front ROI with 3 s lag	3.660e-08	0.095856	90.18	no
Lag sensitivity front ROI with 4 s lag	3.629e-08	0.102262	90.01	no
Lag sensitivity front ROI with 5 s lag	3.610e-08	0.107330	89.52	no
Capped-window sensitivity front ROI	2.146e-08	0.249238	12.00	no
Response-time sensitivity front ROI	3.775e-08	0.089589	90.01	no

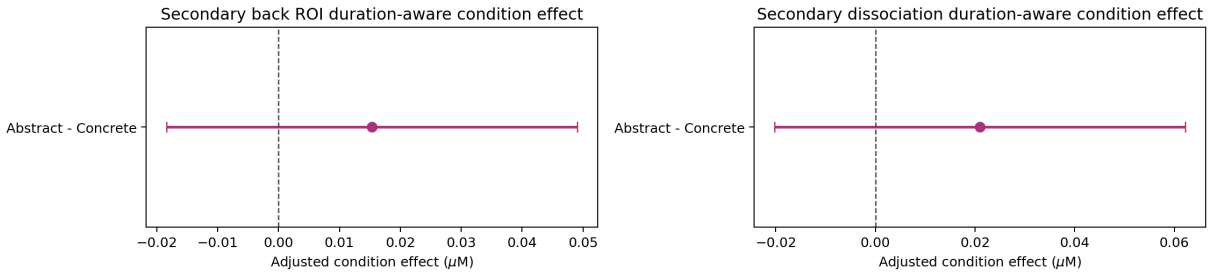

Figure 4: Secondary back ROI and dissociation condition-effect plots.

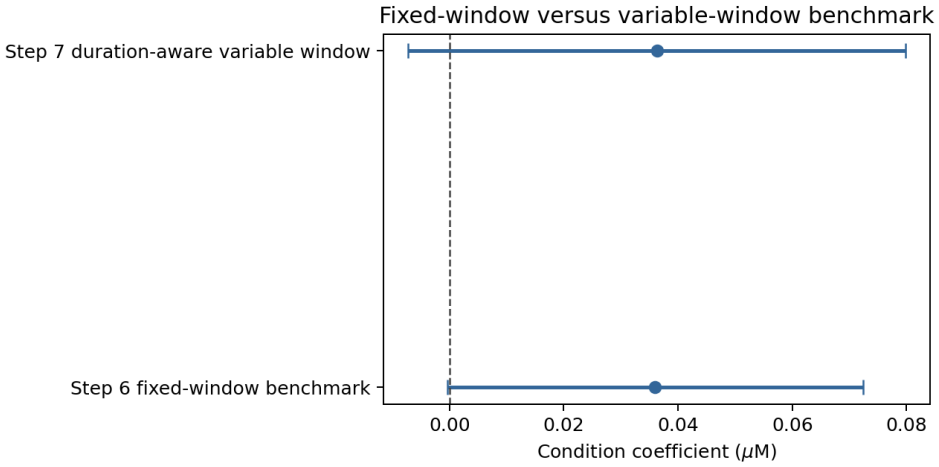

Figure 5: Direct benchmark comparison of the Step 6 fixed-window coefficient and the Step 7 duration-aware coefficient.

Final Conclusion

In the duration-aware mixed-effects analysis, the abstract-versus-concrete effect in the left frontal ROI was not statistically significant after adjusting for log word count and ENEM item correctness (beta=3.629e-08, SE=2.221e-08, p=0.102262, 95